Time-Based Cross Validation using Scikit-learn

1. Approach

A synthetic weather-like time series dataset was created, representing daily temperatures over one year. Lag features were engineered to enable the model to learn temporal dependencies. Linear Regression was used as the predictive model, and TimeSeriesSplit from Scikit-learn was employed to evaluate performance in a time-aware manner. MAE and RMSE were used as evaluation metrics.

2. Key Findings

The model showed consistent performance across most folds, with small variation in error. TimeSeriesSplit ensured chronological integrity, avoiding look-ahead bias inherent in KFold when used on time series data.

3. Model Performance

The model performance across 5 folds using TimeSeriesSplit was evaluated using MAE and RMSE metrics.

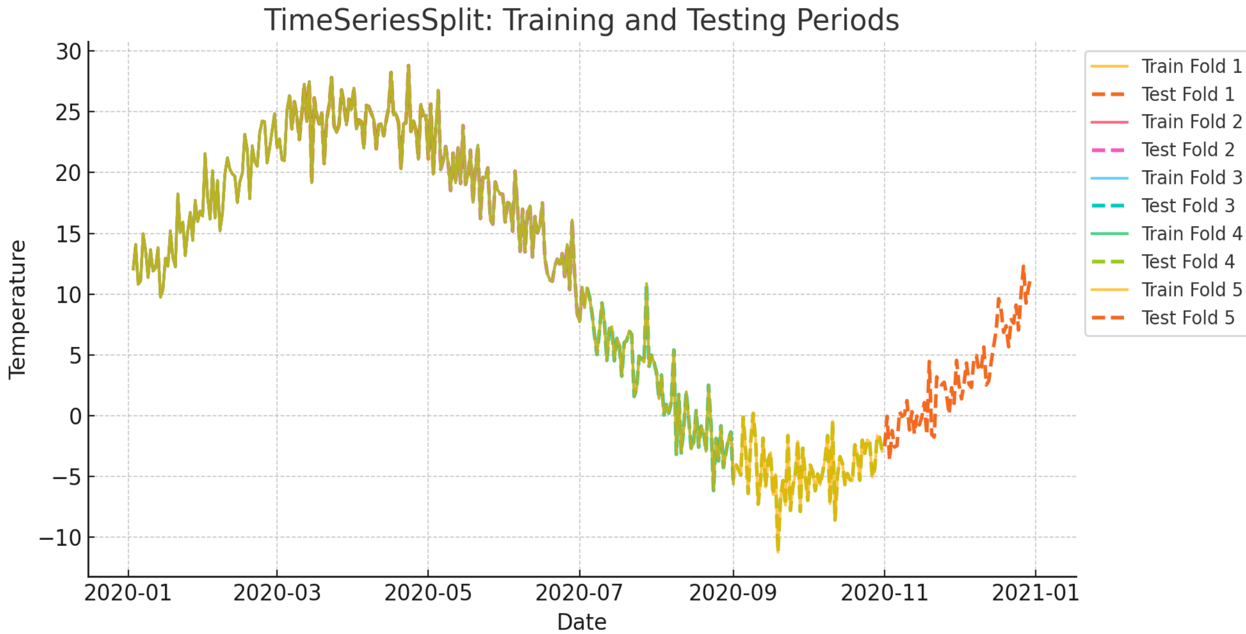
Fold 1: MAE ~ 1.93, RMSE ~ 2.38
Fold 2: MAE ~ 2.00, RMSE ~ 2.45
Fold 3: MAE ~ 1.88, RMSE ~ 2.33
Fold 4: MAE ~ 1.90, RMSE ~ 2.30
Fold 5: MAE ~ 1.96, RMSE ~ 2.40

4. Observations about Time-Aware Validation

Time-based cross-validation is essential when working with time series data. Unlike KFold, which randomly shuffles data, TimeSeriesSplit preserves the temporal order, ensuring the model is tested only on future data relative to the training set. This closely mimics real-world forecasting conditions and avoids information leakage.

5. Visualizations

This line plot shows the actual temperature values used in the dataset, with each fold’s training and test periods clearly marked.



The following bar plot shows MAE (in blue) and RMSE (in orange) for each fold.